ECON 0150 | Homework 4.2

Due: Friday, Nov. 13 at 5PM

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely. You are welcomed and encouraged to work in groups so long as your work is your own. Submit your figures and answers to Gradescope.

Q1. Mental Health and Income Group

The 2011 Behavioral Risk Factor Surveillance System (BRFSS) contains individual-level data with the following variables:

- `MENTHLTH` – days of poor mental health in the past 30 days
- `INCOME2` – income category (1 = less than \$10,000, …, 8 = \$75,000 or more)

The Homework 4.2 Notebook loads the data and, as in Exercise 4.2, drops respondents with a missing income or mental health response. Create a binary variable `under_10k` equal to 1 if `INCOME2 == 1` and 0 otherwise. Then estimate the following linear model:

$$\texttt{MENTHLTH}_i = \beta_0 + \beta_1 \cdot \texttt{under_10k}_i + \varepsilon_i \tag{1}$$

a) Report $\hat{\beta}_0$. What does it represent in context?

b) Report $\hat{\beta}_1$. What does it represent in context?

Q2. Reading Regression Output

A researcher wants to test whether having a college degree is associated with higher hourly wages. They code a binary variable `college` equal to 1 if the worker has a college degree and 0 otherwise, then estimate the following model:

$$\texttt{wage}_i = \beta_0 + \beta_1 \cdot \texttt{college}_i + \varepsilon_i \tag{2}$$

Their regression output shows:

```
               coef  std err        t    P>|t|   [0.025   0.975]
------------------------------------------------------------------
Intercept     18.42    0.310   59.419    0.000    17.81    19.03
college        9.65    0.440   21.932    0.000     8.79    10.51
------------------------------------------------------------------
```

a) Interpret the intercept coefficient (18.42) in context.

b) Interpret the slope coefficient (9.65) in context.

c) What is the null hypothesis being tested for the `college` coefficient?

d) Interpret the p-value in words.